

From Circumference-to-Area Ratios to Dimensional Scaling

A Geometric Path from Circles to n -Balls and Fractal Dimension

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Research-style mathematical exposition

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Origin of the investigation

The investigation began with the elementary observation $C/A = 2/r$ for a circle, together with the special numerical case $r = 2$, and was progressively generalized through sectors, spheres, n -dimensional balls, scaling laws, high-dimensional concentration, arbitrary similar shapes, and non-integer fractal dimension.

Abstract

This paper develops a connected mathematical exposition beginning from the ratio of the circumference C of a circle to its area A . For a circle of radius $r > 0$, $C/A = 2/r$. Although elementary, this identity exposes a broader scaling structure: boundary measures scale one power of length below interior measures. The argument is extended first to circular sectors, then to spheres and n -dimensional balls, yielding $S_{n-1}/V_n = n/r$. The relation is interpreted through logarithmic differentiation, giving $n = d \ln V / d \ln r$, and is then separated into its sphere-specific and shape-independent components. Uniformly similar shapes obey $B/V = K/L$, where K is a dimensionless shape factor and L a characteristic length. The high-dimensional behavior of n -balls is examined, including the concentration of volume near the outer boundary. Finally, the logarithmic scaling viewpoint is generalized to non-integer dimensions and illustrated with the Koch curve and Sierpiński constructions. The paper emphasizes the distinction between an exploratory derivation and established mathematical results: the formulas developed here are known results in geometry and fractal analysis, while the contribution of this exposition is the continuous conceptual route from an elementary circle ratio to dimensional scaling.

1. Starting Point: The Circle

Let a circle have radius r . Under the standard geometric interpretation, r is a magnitude and therefore $r \geq 0$, consistent with $r = \sqrt{x^2 + y^2}$. For a non-degenerate circle we take $r > 0$. A complete rotation corresponds to $\theta = 2\pi$ radians.

$$C = r\theta = 2\pi r$$
$$A = \pi r^2$$

Dividing circumference by area gives

$$C/A = (2\pi r)/(\pi r^2) = 2/r, \quad r > 0.$$

This is the central observation from which the rest of the investigation proceeds. The ratio is not dimensionless: circumference has units of length and area has units of length squared, so C/A has units of inverse length. Consequently, statements such as $C = A$ at a particular numerical radius mean equality of numerical values after a unit convention has been chosen, not equality of physical dimensions.

1.1 The special numerical radius $r = 2$

$$C(2) = 2\pi(2) = 4\pi$$
$$A(2) = \pi(2)^2 = 4\pi$$

Thus $r = 2$ is the unique positive radius for which the numerical values of circumference and area agree. Equivalently, $C/A = 1$ implies $2/r = 1$ and hence $r = 2$.

1.2 Limiting behavior

The ratio must be distinguished from its numerator and denominator. As r approaches infinity, both circumference and area diverge, but area grows quadratically while circumference grows linearly.

$$\lim_{r \rightarrow \infty} C = \infty, \quad \lim_{r \rightarrow \infty} A = \infty, \quad \lim_{r \rightarrow \infty} C/A = 0.$$

Thus $C/A \rightarrow 0$ does not mean $C \rightarrow 0$. It means circumference becomes small relative to area. Similarly, as r approaches zero from the positive side, both C and A approach zero, while area vanishes faster.

$$\lim_{r \rightarrow 0^+} C = 0, \quad \lim_{r \rightarrow 0^+} A = 0, \quad \lim_{r \rightarrow 0^+} C/A = \infty.$$

At $r = 0$ itself the quotient is $0/0$ and is undefined; only the right-hand limit diverges.

2. Removing the Full-Rotation Restriction

The full-circle condition $\theta = 2\pi$ is not necessary for the ratio. Consider a circular sector with central angle θ in radians, where $\theta > 0$.

$$L = r\theta$$

$$A_{\text{sector}} = \frac{1}{2}r^2\theta$$

$$L/A_{\text{sector}} = (r\theta)/(\frac{1}{2}r^2\theta) = 2/r.$$

The angular factor cancels. Therefore every sector of a fixed radius has the same arc-length-to-sector-area ratio. The circle is simply the special case $\theta = 2\pi$. This reveals that the inverse-radius behavior is caused by radial scaling rather than by completion of a full rotation.

3. The Three-Dimensional Analogue

For a sphere of radius r , surface area S and enclosed volume V are

$$S = 4\pi r^2$$

$$V = (4/3)\pi r^3.$$

Their ratio is

$$S/V = (4\pi r^2)/((4/3)\pi r^3) = 3/r.$$

The circle supplied the coefficient 2, while the sphere supplies the coefficient 3. This suggests that the coefficient is connected to dimension.

4. Generalization to an n-Dimensional Ball

The volume of an n -dimensional Euclidean ball of radius r can be written

$$V_n(r) = \kappa_n r^n, \quad \kappa_n = \pi^{(n/2)} / \Gamma(n/2 + 1).$$

The boundary of an n -ball is an $(n-1)$ -sphere. Its measure is obtained by differentiating the ball volume with respect to radius:

$$S_{n-1}(r) = dV_n/dr = n\kappa_n r^{n-1}.$$

Dividing gives the general identity

$$S_{n-1}/V_n = n/r.$$

The constants κ_n cancel completely. The result therefore depends only on dimension and radius.

Dimension n	Interior measure	Boundary measure	Boundary / interior
1	$2r$	2	$1/r$
2	πr^2	$2\pi r$	$2/r$
3	$(4/3)\pi r^3$	$4\pi r^2$	$3/r$
4	$(\pi^2/2)r^4$	$2\pi^2 r^3$	$4/r$
n	$\kappa_n r^n$	$n\kappa_n r^{n-1}$	n/r

4.1 Two special radii

For the function $f_n(r) = n/r$, two different special radii arise. First, boundary and interior measures have equal numerical values when $f_n(r) = 1$:

$$n/r = 1 \Rightarrow r = n.$$

The original circle result $r = 2$ is therefore the $n = 2$ member of the sequence $r = n$. Second, a fixed point of f_n satisfies $f_n(r) = r$:

$$n/r = r \Rightarrow r^2 = n \Rightarrow r = \sqrt{n}.$$

These conditions answer different questions and should not be conflated.

5. Differential and Logarithmic Interpretation

The n -ball formula can be expressed more fundamentally through logarithmic differentiation. Starting from $V_n = \kappa_n r^n$,

$$\ln V_n = \ln \kappa_n + n \ln r.$$

Differentiation with respect to $\ln r$ gives

$$d \ln V_n / d \ln r = n.$$

Using $dV_n/dr = S_{n-1}$, this becomes

$$n = (r/V_n)(dV_n/dr) = rS_{n-1}/V_n.$$

Thus, for n -balls, dimension can be recovered from the radial scaling rate:

$$n = rS_{n-1}/V_n.$$

For an infinitesimal radial increase dr ,

$$dV_n/V_n = n(dr/r).$$

Hence a small fractional change in radius produces approximately n times that fractional change in n -dimensional volume. This is the local scaling interpretation of dimension.

6. High-Dimensional Geometry

Holding r fixed while increasing n gives $S_{n-1}/V_n = n/r$, so the boundary-to-volume ratio grows without bound as $n \rightarrow \infty$. This statement alone does not imply that either boundary measure or volume individually diverges.

For the unit n -ball,

$$V_n(1) = \pi^{(n/2)} / \Gamma(n/2 + 1).$$

The sequence initially increases, reaches its maximum at dimension 5, and then tends to zero as n grows. Thus a vanishing high-dimensional unit-ball volume can coexist with an increasing boundary-to-volume ratio.

6.1 Concentration near the boundary

Let an inner ball have radius αR inside an n -ball of radius R , with $0 < \alpha < 1$. Since volume scales as the n th power of radius,

$$V_n(\alpha R)/V_n(R) = \alpha^n.$$

For $\alpha = 0.9$, the fraction of volume inside 90% of the radius is 0.9^n . At $n = 100$ this is approximately 2.66×10^{-5} , so about 99.9973% of the volume lies in the outermost 10% of the radius. In this precise sense, high-dimensional ball volume becomes concentrated near its boundary.

7. What Is Special to Balls, and What Is General?

The exact coefficient n/r depends on using the radius of a ball. The inverse-length scaling, however, is far more general. Consider any fixed shape uniformly scaled by a factor λ . Its n -dimensional interior measure scales as λ^n , while its $(n-1)$ -dimensional boundary measure scales as λ^{n-1} .

$$V(\lambda) = V_0 \lambda^n, \quad B(\lambda) = B_0 \lambda^{n-1}.$$

$$B(\lambda)/V(\lambda) = (B_0/V_0)(1/\lambda).$$

If L denotes a characteristic length, the relation may be written

$$B/V = K/L,$$

where K is a dimensionless constant determined by the chosen shape and definition of L . For an n -ball with $L = r$, $K = n$. For an n -dimensional hypercube with side length s , $V = s^n$ and $B = 2ns^{n-1}$, giving $B/V = 2n/s$.

Therefore the universal structural statement for geometrically similar objects is not that B/V always equals n/r , but that boundary-to-content ratios scale inversely with overall size.

8. Isoperimetric Efficiency

Boundary-to-content ratios also lead naturally to the isoperimetric problem: for a fixed interior measure, which shape requires the least boundary? In two dimensions the circle minimizes perimeter for fixed area; in three dimensions the sphere minimizes surface area for fixed volume; and the n -ball is the corresponding optimizer in Euclidean n -space.

For example, a square and circle with the same area A have

$$P_{\text{square}} = 4\sqrt{A}$$

$$C_{\text{circle}} = 2\sqrt{(\pi A)}$$

$$C_{\text{circle}}/P_{\text{square}} = \sqrt{\pi}/2 \approx 0.886.$$

Thus the circle uses only about 88.6% of the boundary length required by a square enclosing the same area. The boundary-to-content ratio can therefore be interpreted as a geometric efficiency measure, provided comparisons are made at a common scale or common interior measure.

9. From Integer Dimension to Scaling Dimension

The logarithmic form suggests a broader definition. Suppose a measurable quantity $M(r)$ follows a power law

$$M(r) = Kr^D.$$

Then

$$D = d \ln M / d \ln r = (r/M)(dM/dr).$$

For ordinary Euclidean balls, M is volume and $D = n$. But the exponent need not be an integer. This is the bridge from ordinary geometric dimension to scaling dimensions used in fractal geometry.

Care is required here: for a general fractal, dM/dr should not automatically be interpreted as a conventional geometric boundary. The robust statement is the logarithmic scaling law itself.

10. Self-Similar Fractals

For an exactly self-similar set consisting of N copies of itself, each reduced by a linear factor $1/s$, the similarity dimension D satisfies

$$N = s^D$$

$$D = \ln N / \ln s.$$

Equivalently, when a structure is measured using a resolution ϵ that tends to zero, a covering number often behaves as $N(\epsilon) \sim \epsilon^{-D}$, leading to

$$D = -d \ln N(\epsilon) / d \ln \epsilon.$$

This form makes explicit why the sign changes: finer resolution means smaller ϵ but a larger number of required covering elements.

10.1 Koch curve

Each segment of the Koch construction is replaced by four segments, each one third as long. Thus $N = 4$ and $s = 3$:

$$D_{\text{Koch}} = \ln 4 / \ln 3 \approx 1.26186.$$

The value lies strictly between 1 and 2. The curve is more intricate than a smooth one-dimensional line but does not fill a two-dimensional region.

10.2 Sierpiński triangle

The Sierpiński triangle consists of three self-similar copies, each scaled by one half:

$$D_{\text{Sierpiński}} = \ln 3 / \ln 2 \approx 1.58496.$$

Again, the scaling dimension lies between one and two even though the construction is embedded in the plane.

10.3 Sierpiński tetrahedron

A Sierpiński tetrahedron consists of four copies scaled by one half:

$$D = \ln 4 / \ln 2 = 2.$$

It is embedded in three-dimensional Euclidean space but has similarity dimension 2. This demonstrates the distinction between embedding dimension and scaling dimension.

11. Unified Derivation

The entire development can be summarized as a sequence of increasingly general statements:

- Circle: $C/A = 2/r$.
- Sector: arc length / sector area = $2/r$, independent of θ .
- Sphere: $S/V = 3/r$.
- n -ball: $S_{n-1}/V_n = n/r$.
- Scaling inversion for n -balls: $n = rS_{n-1}/V_n$.
- Uniform similarity scaling: $B/V = K/L$.
- General power-law scaling: $D = d \ln M / d \ln r$.
- Exact self-similarity: $D = \ln N / \ln s$.

12. Interpretation and Limits of the Argument

The progression above should be interpreted as an exploratory mathematical route, not as a claim that the established formulas were newly discovered. The n -ball volume formula, the boundary derivative identity, logarithmic scaling exponents, isoperimetric inequalities, and fractal similarity dimensions are standard results. What is distinctive in this exposition is the route by which an elementary comparison of circumference and area naturally motivates them.

Several conceptual cautions are essential. First, quantities of different physical dimensions cannot literally be equated without a unit convention; $C = A$ at $r = 2$ refers to numerical equality. Second, a quotient approaching zero does not force its numerator to approach zero. Third, the identity $S_{n-1} = dV_n/dr$ is particularly natural for balls and parallel radial shells; it must not be assumed for arbitrary fractal boundaries. Fourth, a scaling dimension and the dimension of the ambient space are distinct concepts.

13. Conclusion

Beginning with $C/A = 2/r$, a simple circle calculation reveals a recurring geometric principle: when an interior measure scales as the n th power of length, its associated codimension-one boundary scales one power lower. For Euclidean n -balls this produces the exact relation $S_{n-1}/V_n = n/r$ and the logarithmic identity $n = d \ln V_n / d \ln r$. Uniform scaling shows that the inverse-size behavior extends beyond spheres, with a shape-dependent coefficient. High-dimensional analysis demonstrates how this scaling contributes to concentration near the boundary, while the logarithmic formulation naturally extends from integer Euclidean dimensions to non-integer fractal dimensions. The path from a circle to a Koch curve is therefore not a collection of unrelated formulas: it is a continuous study of how geometric measure changes with scale.

Appendix A. Compact Formula Sheet

Object / concept	Relation
Circle	$C = 2\pi r$; $A = \pi r^2$; $C/A = 2/r$
Sector	$L = r\theta$; $A = \frac{1}{2}r^2\theta$; $L/A = 2/r$
Sphere	$S = 4\pi r^2$; $V = (4/3)\pi r^3$; $S/V = 3/r$
n -ball	$V_n = \kappa_n r^n$; $S_{n-1} = n\kappa_n r^{n-1}$; $S_{n-1}/V_n = n/r$
n -ball constant	$\kappa_n = \pi^{(n/2)} / \Gamma(n/2 + 1)$
Scaling dimension	$D = d \ln M / d \ln r$
Similarity dimension	$D = \ln N / \ln s$
Resolution form	$D = -d \ln N(\epsilon) / d \ln \epsilon$

References and Mathematical Context

This exposition uses standard results from Euclidean geometry, multivariable calculus, geometric measure scaling, the isoperimetric principle, and elementary fractal geometry. For formal publication, a subsequent literature-review version should add bibliographic citations to authoritative texts on n -dimensional ball volumes, isoperimetric inequalities, and Hausdorff/similarity dimension. No claim of priority is made for the established mathematical identities presented here.